EEE 101 MASTER FORMULA SHEET

1. Basic Concepts, Laws, Dividers & Y-Δ Transformations

Charge, Current, Voltage, Power & Energy

$$i(t) = \frac{dq}{dt}, \qquad q(t) = \int_{t_0}^{t} i(\tau)\,d\tau + q(t_0), \qquad v_{ab} = \frac{dw}{dq} = v_a - v_b, \qquad p(t) = \frac{dw}{dt} = v(t)i(t)$$

$$w(t) = \int_{t_0}^{t} p(\tau)\,d\tau = \int_{t_0}^{t} v(\tau)i(\tau)\,d\tau, \qquad \sum_{\text{all branches}} p_k(t) = 0 \implies \sum P_{\text{absorbed}} = \sum P_{\text{supplied}}$$

Practical Sources: $v_L = v_s - i_L R_s = \dfrac{R_L}{R_s + R_L} v_s, \qquad i_L = i_s - \dfrac{v_L}{R_p} = \dfrac{R_p}{R_p + R_L} i_s, \qquad v_s = i_s R_s$

Ohm's Law, Static & Dynamic Resistance, Topology & Dividers

$$R = \rho\frac{l}{A}, \qquad G = \frac{1}{R} = \sigma\frac{A}{l}, \qquad v = iR = \frac{i}{G}, \qquad p = vi = i^2R = \frac{v^2}{R} = v^2G = \frac{i^2}{G}$$

$$R_{\text{static}} = R_{\text{DC}} = \frac{V_Q}{I_Q}, \qquad r_{\text{dynamic}} = r_d = \left.\frac{dv}{di}\right|_Q, \qquad R(T) = R_0\left[1 + \alpha_0(T - T_0)\right]$$

$$b = l + n - 1, \qquad \text{KCL: } \sum_{k=1}^{N} i_k = 0 \quad \left(\sum i_{\text{in}} = \sum i_{\text{out}}\right), \qquad \text{KVL: } \sum_{m=1}^{M} v_m = 0 \quad \left(\sum v_{\text{drops}} = \sum$$

$$R_{\text{series}} = \sum_{k=1}^{N} R_k, \qquad \frac{1}{R_{\text{parallel}}} = \sum_{k=1}^{N} \frac{1}{R_k} \implies G_{\text{eq}} = \sum_{k=1}^{N} G_k, \qquad R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

Voltage Divider: $v_k = \dfrac{R_k}{R_1 + R_2 + \cdots + R_N} v_s, \qquad$ Current Divider: $i_k = \dfrac{G_k}{G_1 + G_2 + \cdots + G_N} i_s, \quad i_1 = \dfrac{R_2}{R_1 + R_2}$

Ammeter Shunt: $R_{\text{sh}} = \dfrac{I_{\text{fs}} R_m}{I - I_{\text{fs}}}, \qquad$ Voltmeter Multiplier: $R_{\text{mult}} = \dfrac{V - I_{\text{fs}} R_m}{I_{\text{fs}}}, \qquad$ Half-Deflection: $R_x = R -$

$\Delta \leftrightarrow$ Y Transformations & Special Resistive Networks

$$\Delta \to \text{Y}: \quad R_1 = \frac{R_b R_c}{R_a + R_b + R_c}, \qquad R_2 = \frac{R_c R_a}{R_a + R_b + R_c}, \qquad R_3 = \frac{R_a R_b}{R_a + R_b + R_c}, \qquad R_Y =$$

$$\text{Y} \to \Delta: \quad R_a = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_1}, \qquad R_b = \frac{R_1R_2 + R_2R_3 + R_3R_1}{R_2}, \qquad R_c = \frac{R_1R_2 +$$

Wheatstone Bridge Balance: $\dfrac{R_1}{R_2} = \dfrac{R_3}{R_x} \implies R_x = \dfrac{R_2}{R_1} R_3 \quad (i_g = 0,\ v_g = 0)$

Infinite Ladder ($R_{\text{series}}, R_{\text{shunt}}$): $R_{\text{in}} = R_{\text{ser}} + \dfrac{R_{\text{sh}} R_{\text{in}}}{R_{\text{sh}} + R_{\text{in}}} \implies R_{\text{in}} = \dfrac{R_{\text{ser}} + \sqrt{R_{\text{ser}}^2 + 4R_{\text{ser}}R_{\text{sh}}}}{2}$

Sliding Load ($v_1, v_2, r\ \Omega$/m): $v(x) = \dfrac{v_1 RL + R(v_2 - v_1)x}{RL + 2rLx - 2rx^2}, \qquad x_{\text{min}} = \dfrac{L}{v_2 - v_1}\left[-v_1 \pm \sqrt{v_1 v_2 - \dfrac{R}{2rL}(v_1 - v_2)^2}\right]$

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2. Methods of Analysis & Network Theorems

Nodal/Mesh Matrix Inspection, Linearity, Superposition, Thevenin & Norton

Nodal Matrix (No V-Sources): $[G][V] = [I], \qquad G_{kk} = \sum$ conductances at node $k, \qquad G_{kj} = G_{jk} = -\sum G_{\text{betw}}$

Mesh Matrix (No I-Sources): $[R][I] = [V], \qquad R_{kk} = \sum$ resistances in mesh $k, \qquad R_{kj} = R_{jk} = -\sum R_{\text{shared b}}$

Linearity & Superposition: $f(ax_1 + bx_2) = af(x_1) + bf(x_2), \qquad v = \sum_{k=1}^{N} v_k, \qquad i = \sum_{k=1}^{N} i_k, \qquad p = i^2R = \left(\sum_{k=1}^{N}\right.$

Thevenin & Norton: $V_{\text{Th}} = v_{\text{oc}}, \qquad I_N = i_{\text{sc}}, \qquad R_{\text{Th}} = R_N = \dfrac{v_{\text{oc}}}{i_{\text{sc}}} = \dfrac{V_{\text{test}}}{I_{\text{test}}}, \qquad v_L = V_{\text{Th}} - R_{\text{Th}} i_L$

Maximum Power Transfer, Millman's & Reciprocity Theorems

DC Max Power Transfer: $R_L = R_{\text{Th}}, \qquad P_{\text{max}} = \dfrac{V_{\text{Th}}^2}{4R_{\text{Th}}} = \dfrac{I_N^2 R_N}{4}, \qquad \eta_{\text{max power}} = \dfrac{R_L}{R_{\text{Th}} + R_L} = 50\%$

Millman's Theorem: $V_{\text{eq}} = \dfrac{\sum_{k=1}^{n} \frac{\pm V_k}{R_k}}{\sum_{k=1}^{n} \frac{1}{R_k}} = \dfrac{\sum_{k=1}^{n} \pm G_k V_k}{\sum_{k=1}^{n} G_k}, \qquad R_{\text{eq}} = \dfrac{1}{\sum_{k=1}^{n} \frac{1}{R_k}} = \dfrac{1}{\sum_{k=1}^{n} G_k}$

Reciprocity Theorem: $\left.\dfrac{I_2}{V_1}\right|_{V_2=0} = \left.\dfrac{I_1}{V_2}\right|_{V_1=0} = y_{21} = y_{12}, \qquad \left.\dfrac{V_2}{I_1}\right|_{I_2=0} = \left.\dfrac{V_1}{I_2}\right|_{I_1=0} = z_{21} = z_{12}$

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3. Energy Storage Elements, Singularity Functions & Transients (RC, RL, RLC)

Capacitors, Inductors & Coupled Inductance

Capacitor (C): $q = Cv, \qquad i = C\dfrac{dv}{dt}, \qquad v(t) = \dfrac{1}{C}\int_{t_0}^{t} i(\tau)\, d\tau + v(t_0), \qquad w_C = \dfrac{1}{2}Cv^2 = \dfrac{q^2}{2C}, \qquad v_C(0^+) = v_C(0^-$

$C_{\text{parallel}} = \sum_{k=1}^{N} C_k, \qquad \dfrac{1}{C_{\text{series}}} = \sum_{k=1}^{N} \dfrac{1}{C_k}, \qquad v_{C1} = \dfrac{C_2}{C_1 + C_2} v_s$ (Series C Voltage Divider)

Inductor (L): $v = L\dfrac{di}{dt}, \qquad i(t) = \dfrac{1}{L}\int_{t_0}^{t} v(\tau)\, d\tau + i(t_0), \qquad w_L = \dfrac{1}{2}Li^2, \qquad i_L(0^+) = i_L(0^-)$

$L_{\text{series}} = \sum_{k=1}^{N} L_k, \qquad \dfrac{1}{L_{\text{parallel}}} = \sum_{k=1}^{N} \dfrac{1}{L_k}, \qquad i_{L1} = \dfrac{L_2}{L_1 + L_2} i_s$ (Parallel L Current Divider)

Duality Pairs: $v \leftrightarrow i, \qquad R \leftrightarrow G = \dfrac{1}{R}, \qquad L \leftrightarrow C, \qquad q \leftrightarrow \phi,$ Series $\leftrightarrow$ Parallel, KVL $\leftrightarrow$ KCL

Singularity Functions

$$u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases}, \qquad u(-t) = 1 - u(t) = \begin{cases} 1, & t < 0 \\ 0, & t > 0 \end{cases}, \qquad \text{Gate}[t_1, t_2] = u(t - t_1) - u(t - t_2)$$

$$\delta(t - t_0) = \frac{d}{dt}u(t - t_0), \qquad \int_{-\infty}^{\infty} \delta(t - t_0)\, dt = 1, \qquad \int_a^b f(t)\delta(t - t_0)\, dt = \begin{cases} f(t_0), & a < t_0 < b \\ 0, & \text{otherwise} \end{cases}$$

$$r(t - t_0) = \int_{-\infty}^{t} u(\tau - t_0)\, d\tau = (t - t_0)u(t - t_0), \qquad \frac{d}{dt}r(t - t_0) = u(t - t_0)$$

First-Order RC and RL Transient Circuits

$$\text{Time Constant:} \quad \tau_{RC} = R_{\text{Th}}C, \qquad \tau_{RL} = \frac{L}{R_{\text{Th}}}, \qquad x(t) = x(\infty) + \left[x(t_0^+) - x(\infty)\right] e^{-(t-t_0)/\tau} \quad (t \geq t_0)$$

$$\text{Source-Free } (x(\infty) = 0): \quad v_C(t) = V_0 e^{-t/(RC)}, \qquad i_L(t) = I_0 e^{-(R/L)t}, \qquad \tau = -\frac{x(0)}{dx(0)/dt}$$

$$\text{Energy Delivery:} \quad w_R(t) = W_0\left(1 - e^{-2t/\tau}\right), \qquad \frac{w_R(t)}{W_0} = \eta \implies \frac{t}{\tau} = -\frac{1}{2}\ln(1-\eta) \quad (\eta = 0.95 \implies t \approx 1.5$$

Second-Order Series & Parallel RLC Transient Circuits

$$s^2 + 2\alpha s + \omega_0^2 = 0, \qquad s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}, \qquad \omega_0 = \frac{1}{\sqrt{LC}}, \qquad \omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

$$\text{Series } RLC: \quad \alpha = \frac{R}{2L}, \qquad \frac{dv_C(0^+)}{dt} = \frac{i_C(0^+)}{C}, \qquad \frac{di_L(0^+)}{dt} = \frac{v_L(0^+)}{L} = \frac{V_s - v_R(0^+) - v_C(0^+)}{L}$$

$$\text{Parallel } RLC: \quad \alpha = \frac{1}{2RC}, \qquad \frac{dv_C(0^+)}{dt} = \frac{I_s - i_R(0^+) - i_L(0^+)}{C}, \qquad \frac{di_L(0^+)}{dt} = \frac{v_C(0^+)}{L}$$

1. Overdamped ($\alpha > \omega_0$): $x(t) = x(\infty) + A_1 e^{s_1 t} + A_2 e^{s_2 t}, \qquad t_{\text{peak}} = \dfrac{\ln(s_2/s_1)}{s_1 - s_2}$ (when $A_2 = -A_1$)
2. Critically Damped ($\alpha = \omega_0$): $x(t) = x(\infty) + (A_1 + A_2 t)e^{-\alpha t}, \qquad t_{\text{peak}} = \dfrac{1}{\alpha} - \dfrac{A_1}{A_2}$
3. Underdamped ($\alpha < \omega_0$): $x(t) = x(\infty) + e^{-\alpha t}\left(A_1 \cos \omega_d t + A_2 \sin \omega_d t\right)$

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4. Magnetic Circuits

$$B = \frac{\Phi}{A} \quad (\text{T or Wb/m}^2), \qquad \mu = \mu_0\mu_r = \frac{B}{H}, \qquad \mu_0 = 4\pi \times 10^{-7}\ \text{H/m}, \qquad \mathcal{F} = NI = Hl =$$

$$H = \frac{NI}{l} = \frac{B}{\mu} \quad (\text{At/m}), \qquad H_g = \frac{B_g}{\mu_0} = \frac{B_g}{4\pi \times 10^{-7}} \approx 7.958 \times 10^5 B_g\ \text{At/m}, \qquad \mathcal{R} = \frac{l}{\mu A}$$

$$\text{Ampere's Circuital Law:} \quad \oint \vec{H} \cdot d\vec{l} = \sum_{k=1}^{m} H_k l_k = \sum NI \quad \Longleftrightarrow \quad \sum \mathcal{F} = \sum \Phi_k \mathcal{R}_k = \sum H_k l_k$$

$$\text{Flux Node Law:} \quad \sum \Phi_{\text{in}} = \sum \Phi_{\text{out}}, \qquad F_{\text{air gap}} = \frac{B_g^2 A_g}{2\mu_0}, \qquad 1\ \text{in.} = 0.0254\ \text{m}, \qquad 1\ \text{in.}^2 = 6.4516 \times 10^{-4}$$

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5. AC Steady-State Analysis, Periodic Waveforms & AC Power

Periodic Waveforms: Average & RMS (Effective) Values

$$V_{\text{avg}} = \frac{1}{T}\int_0^T v(t)\,dt, \qquad V_{\text{rms}} = V_{\text{eff}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}, \qquad \text{For}$$

$$\text{Segment Integrals of } \int_{t_1}^{t_2} v^2(t)\,dt \text{ over } \Delta t: \quad \text{Constant } V_0 \implies V_0^2 \Delta t, \qquad \text{Ramp } 0 \to V_m \implies \frac{1}{3}V_m^2 \Delta t$$

$$\text{General Ramp } V_1 \to V_2 \implies \frac{1}{3}(V_1^2 + V_1V_2 + V_2^2)\Delta t, \quad \text{Sine Arch } V_m \sin \omega t \implies \frac{1}{2}V_m^2 \Delta t, \quad I_{\text{rms}} = \sqrt{I_0^2 + \frac{1}{2}\sum_{k=1}^{N} I_{mk}^2}$$

Phasors, Impedance, Admittance & Resonance

$$v(t) = V_m \cos(\omega t + \phi) \Longleftrightarrow \mathbf{V} = V_m \angle\phi = V_{\text{rms}} \angle\phi, \qquad \sin(\omega t + \theta) = \cos(\omega t + \theta - 90^\circ)$$

$$\frac{dv}{dt} \Longleftrightarrow j\omega \mathbf{V}, \qquad \int v\,dt \Longleftrightarrow \frac{\mathbf{V}}{j\omega}, \qquad Z_R = R, \qquad Z_L = j\omega L = jX_L, \qquad Z_C = \frac{1}{j\omega C} = -j$$

$$Z = \frac{\mathbf{V}}{\mathbf{I}} = R + jX = |Z| \angle\theta_Z, \qquad Y = \frac{1}{Z} = G + jB = \frac{R}{R^2 + X^2} + j\frac{-X}{R^2 + X^2}$$

Resonance ($\text{Im}(Z) = 0$): $\omega_0 = \dfrac{1}{\sqrt{LC}}, \qquad Q_{\text{series}} = \dfrac{\omega_0 L}{R} = \dfrac{1}{\omega_0 CR}, \qquad Q_{\text{parallel}} = \dfrac{R}{\omega_0 L} = \omega_0 CR, \qquad B = \dfrac{\omega_0}{Q}$

Instantaneous, Average, Reactive & Complex Power, PF Correction & Max Power

$$p(t) = v(t)i(t) = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i) = P + P\cos$$

$$P = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos\theta = I_{\text{rms}}^2 R = \frac{V_{R,\text{rms}}^2}{R} \quad \text{(W)}$$

$$Q = \frac{1}{2} V_m I_m \sin(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \sin\theta = I_{\text{rms}}^2 X = \frac{V_{X,\text{rms}}^2}{X} \quad \text{(VAR : } +Q \text{ Ind}$$

$$\mathbf{S} = P + jQ = \frac{1}{2}\mathbf{V}\mathbf{I}^* = \mathbf{V}_{\text{rms}}\mathbf{I}_{\text{rms}}^* = I_{\text{rms}}^2 Z = \frac{V_{\text{rms}}^2}{Z^*}, \qquad S = |\mathbf{S}| = \sqrt{P^2 + Q^2} =$$

$$\text{pf} = \cos(\theta_v - \theta_i) = \frac{P}{S} = \frac{R}{|Z|} \quad \text{(Lagging if } \theta_i < \theta_v \text{ / Inductive; Leading if } \theta$$

Conservation of AC Power: $\mathbf{S}_T = \sum_{k=1}^{N} \mathbf{S}_k = \sum_{k=1}^{N} P_k + j \sum_{k=1}^{N} Q_k, \qquad Q_k = \pm P_k \tan(\cos^{-1} \text{pf}_k)$

Power Factor Correction ($\theta_1 \to \theta_2$): $Q_C = Q_1 - Q_2 = P_T(\tan\theta_1 - \tan\theta_2) = \omega C V_{\text{rms}}^2 \implies C = \dfrac{P_T(\tan\theta_1 - \tan$

AC Max Power (Complex Z_L): $Z_L = Z_{\text{Th}}^* = R_{\text{Th}} - jX_{\text{Th}}, \qquad P_{\max} = \dfrac{|V_{\text{Th,peak}}|^2}{8R_{\text{Th}}} = \dfrac{|V_{\text{Th,rms}}|^2}{4R_{\text{Th}}}$

AC Max Power (Purely Resistive R_L): $R_L = |Z_{\text{Th}}| = \sqrt{R_{\text{Th}}^2 + X_{\text{Th}}^2}, \qquad P_{\max} = \dfrac{|V_{\text{Th,rms}}|^2}{2\left(|Z_{\text{Th}}| + R_{\text{Th}}\right)}$